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Effect of prenatal exposure to hydroxyprogesterone on steroidogenic enzymes in male rats

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Abstract Pregnant Wistar strain albino rats were administered hydroxyprogesterone and the steroidogenic potential of the testis was analysed in the next generation adult male rats. In utero exposure to supranormal levels (10 mg/kg body weight or 25 mg/kg body weight) of hydroxyprogesterone decreased the activity levels of marker steroidogenic enzymes (3β -hydroxysteroid dehydrogenase and 17β -hydroxysteroid dehydrogenase) of adult (mature) rat testis. Hydroxyprogesterone mediation in the suppression of rat testicular steroidogenesis is suggested, which can be correlated to the observed decrease in male reproductive potential in mammals exposed to female hormones during embryonic development.

Introduction

Deterioration of male reproductive health in humans and wildlife has been reported during the last few decades. A significant decrease in sperm count (113 million/ml vs 66 million/ml) and sperm volume (3.40 ml vs 2.75 ml) in humans has been observed over the period 1938–1990 (Carlsen et al. 1992; Farrow 1994). Over the same period, the incidence of testicular cancer has tripled (Forman and Moller 1994). Incidences of hypospadias and cryptorchidism have also increased during the same time period (see reviews by Kavlock et al. 1996; Toppari et al. 1996). While the reasons for these differences are currently unknown, both clinical and laboratory research suggest that the adverse changes may be interrelated and may have a common origin in fetal life. Exposure of the male fetus to supranormal levels of oestrogens, such as diethylstilbestrol, or oestrogen-like chemicals has been

suggested as a cause of the above-mentioned reproductive defects (Kavlock et al. 1996).

In view of this, an elaborate programme to systematically evaluate the effect of exposure to supranormal levels of hydroxyprogesterone during the prenatal period was undertaken in this laboratory. Hydroxyprogesterone caproate is a synthetic progesterone prescribed to women to prevent early abortion and abnormal uterine bleeding. Hydroxyprogesterone has a short biological half-life and is converted to testosterone in males and oestradiol in females. The present report is the first one from this research, and examines the changes in 3β -hydroxysteroid dehydrogenase (3β -HSD) and 17β -hydroxysteroid dehydrogenase (17β -HSD) activity levels in the testis of adult rats, which were exposed to supranormal levels of hydroxyprogesterone during embryonic stage.

Materials and methods

Albino rats of the Wistar strain were obtained from the Animal Facility, Department of Zoology, S.V. University, Tirupati. The rats were maintained at $23\pm 1^\circ\text{C}$, under a regulated light and dark (12:12 h) schedule and provided with food and water ad libitum. The rat feed was purchased from Kamadhenu Agencies, Bangalore, India. Adult rats (90 days old) were used for experimentation.

Rats were allowed to mate and pregnant rats were separated and injected with hydroxyprogesterone. Hydroxyprogesterone caproate (trade name: Proluton Depot) was purchased from a local drug store. Hydroxyprogesterone caproate (250 mg in 1.0 ml) is available in an oily solution of castor oil IP and benzyl benzoate IP (1:1.7). Pregnant rats were divided into three groups of six animals each. The first group, which served as control, was treated the same as the experimental group but received injections of a mixture of castor oil and benzyl benzoate (1:1.7) in 15 μl volume. Groups 2 and 3 received injections of 5.6 μl (10 mg/kg body weight) or 14 μl (25 mg/kg body weight) hydroxyprogesterone respectively per injection. Injections were administered intraperitoneally to pregnant rats on the first, seventh and fourteenth day of pregnancy. The male pups from these rats were allowed to grow on a normal diet. After 90 days, the male rats (weighing 180 ± 10 g body weight) were sacrificed by cervical dislocation and the testis was isolated. The tissue was homogenized in ice-cold Tris HCl buffer. The microsomal fraction was separated and used as the enzyme source. The activity levels of 3β -HSD (EC 1.1.1.51; conversion of NAD to NADH) and 17β -HSD (EC 1.1.1.64;

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conversion of NADPH to NADP) were measured by the method of Bergmeyer (1974). The enzyme assays were made under the conditions following zero order kinetics after preliminary standardization regarding linearity with respect to time of incubation and enzyme concentration.

The reaction mixture in a volume of 2.0 ml contained 100 μmol of sodium pyrophosphate buffer (pH 9.0), 0.5 μmol of co-factor (NAD for $3\beta\text{-HSD}$, and NADPH for $17\beta\text{-HSD}$), 0.08 μmol of substrate (dehydro-epiandrosterone for $3\beta\text{-HSD}$ and androstenedione for $17\beta\text{-HSD}$) and 20 mg equivalent of homogenate as enzyme source.

The reactions were carried out in a quartz cuvette of 1.0 cm path at $23\pm 1^\circ\text{C}$. The absorbance at 340 nm was measured at 20-s intervals for 5 min in a UV-VIS spectrophotometer (Hitachi model U-2001) against blank controls. The enzyme activities were expressed in μmol of NAD converted to NADH/mg protein/min ($3\beta\text{-HSD}$) and μmol of NADPH converted to NADP/mg protein/min ($17\beta\text{-HSD}$). The data were statistically analysed by Student's *t*-test (Pillai and Sinha 1968).

Results

The results presented in Figs. 1 and 2 indicate the enzyme activity levels in the testis of rats exposed in utero to hydroxyprogesterone. In general, the $3\beta\text{-HSD}$ activity (0.012 ± 0.0018) was greater in the testis than $17\beta\text{-HSD}$ (0.010 ± 0.0014). A significant decrease in the activity levels of $3\beta\text{-HSD}$ ($df=10$; $P=0.0004$) and $17\beta\text{-HSD}$ ($df=10$; $P<0.0001$) was observed in the testis of rats exposed to hydroxyprogesterone. The decrease in the enzyme activity increased with increased dose of hydroxyprogesterone when compared with the controls.

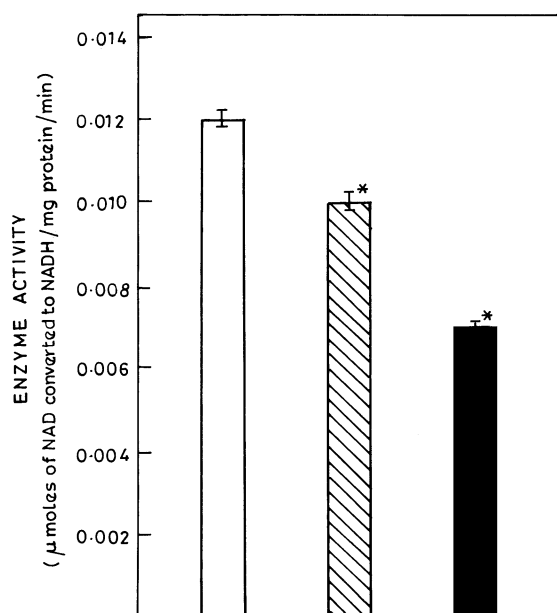


Fig. 1 $3\beta\text{-Hydroxysteroid dehydrogenase}$ activity in the testis of adult rats exposed in utero to various doses of hydroxyprogesterone. Values expressed as μmol of NADH converted to NAD/mg protein per min are means $\pm$ SD of six individuals. Open column normal, hatched column 10 mg hydroxyprogesterone/kg body weight, solid column 25 mg hydroxyprogesterone/kg body weight. Asterisks denote values significantly ($P<0.001$) different from control

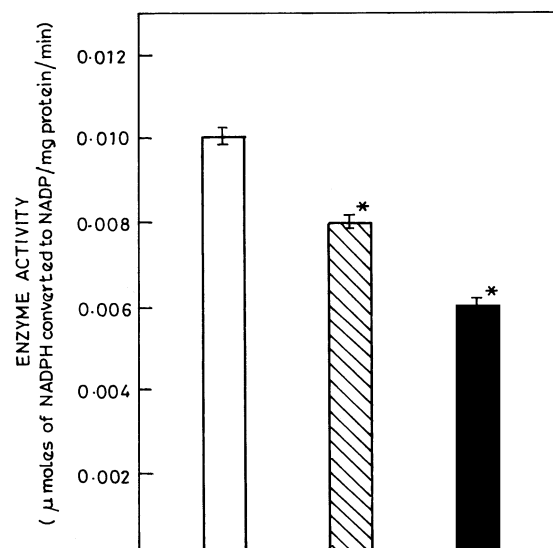


Fig. 2 $17\beta\text{-Hydroxysteroid dehydrogenase}$ activity in the testis of adult rats exposed in utero to various doses of hydroxyprogesterone. Values expressed as μmol of NADP converted to NADPH/mg protein per min are mean $\pm$ SD of six individuals. Open column normal, hatched column 10 mg hydroxyprogesterone/kg body weight, solid column 25 mg hydroxyprogesterone/kg body weight. Asterisks denote values significantly ($P<0.001$) different from control

Discussion

Since the enzymes $3\beta\text{-HSD}$ and $17\beta\text{-HSD}$ are important in the synthesis of most biologically active steroids (Wiebe 1969), determination of the activity levels of these enzymes can be an invaluable aid in subjectively assessing the state of the steroid biosynthesis. The decreased enzyme activity may lead to decreased steroidogenesis in experimental rats. Injection of estradiol- 17β has already been shown to affect $3\beta\text{-HSD}$ and $17\beta\text{-HSD}$ and testosterone production in rats (Chengamma and Reddanna 1987). Decreased levels of serum testosterone were also found in rats exposed in utero to excess hydroxyprogesterone (unpublished results). Results of this experiment indicate that decreased hydroxysteroid dehydrogenase enzyme activity levels in hydroxyprogesterone injected rats, lead to lesser quantities of steroid substrate material to be synthesized.

Maintenance of steroid hormone homeostasis within the organism is critical to ensure normal sexual functions in males and females, including sexual differentiation. Recently, concern has been raised that humans are being exposed to higher concentrations of oestrogenic chemicals in the form of oral contraceptives and hormone injections to programme and protect pregnancy. These substances may be responsible for eliciting adverse effects on development and fertility of the next generation. One such example is diethylstilbestrol (DES), a synthetic oestrogen that was prescribed to more than 5 million pregnant women from the late 1940s to the early 1970s to prevent abortions and pregnancy complications

(Palmlund et al. 1993). Recent clinical observations, including increased incidence of male genital tract malformations (Beard et al. 1984; Matlai and Beral 1985; Ancel et al. 1992), reduction in sperm count (Schumacher et al. 1981; Carlsen et al. 1992), and increased incidence of testicular (Depue et al. 1983; Brown et al. 1986; Quinn and Allen 1995) and prostate cancer (Hendersen et al. 1979; Merrill and Brawley 1997) have been suggested to be linked to exposure to DES in utero. It was also observed that injection of medroxyprogesterone acetate causes increased incidences of uterine and ovarian cancer in humans (Liang et al. 1983). An increased rate of cryptorchidism was reported in sons whose mothers took oral contraceptives in early pregnancy (Rothman and Louik 1978). There are reports indicating that prenatal exposure of male mice to ethinyl oestradiol causes abnormalities in the development of reproductive organs (Yasuda et al. 1988; Thayer et al. 2001), fetal testis differentiation (Yasuda et al. 1985), and Leydig cell hyperplasia (Yasuda et al. 1986).

The increasing incidence of certain human male reproductive tract abnormalities (such as cryptorchidism and hypospadias) have been observed during periods of increased exposure to oestrogenic chemicals (Sharpe and Skakkebaek 1993). Evidence indicates that changes in the concentrations of sex hormones (androgen and oestrogen) result in permanent changes in cell function. For example, the higher circulating levels of oestradiol (about 50 pg/ml) in male mouse fetuses result in an alteration in prostrate size (vom Saal et al. 1997). We have also observed that prenatal exposure to hydroxyprogesterone results in decreased sperm count, sperm viability, sperm motility and circulating testosterone levels in male rats exposed (unpublished results) and suggest that the abnormalities in male reproduction induced by hydroxyprogesterone may be mediated by the synthesis of androgen.

It can be concluded that in utero exposure to supra-normal levels of hydroxyprogesterone suppresses testicular steroidogenesis by decreasing the steroidogenic enzyme activity levels, which in turn may suppress the reproductive activities in the male rats. Of course the rat model and the dose of hydroxyprogesterone used in the present study can constitute a paradigm for comparison to the exposure of fetuses to low doses of similar intracellular activity of xenobiotic persistent compounds man-made and released in to the environment.

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